

quantum safe

$$\vec{c} = A \cdot \vec{m} + \vec{e}$$

where A is a defined matrix, $\vec{e}$ is a random error vector (short with respect to A and $\vec{m}$, $\vec{m}$ is $(\text{ord}(\mathbf{c}), \text{random}, \text{random})$)

we can rewrite this ciphertext equation

$$\vec{c} - A \cdot \vec{m} = \vec{e}$$

we can now see what's the "path" to find, for example, e_1

$$e_1 = c_1 - (m_1 \cdot A_{11} + m_2 \cdot A_{12} + m_3 \cdot A_{13})$$

we can replicate this behavior for all elements of $\vec{e}$:

$$\begin{aligned} &1 \cdot (c_1, c_2, c_3) + \\ &m_1 \cdot (-A_{11}, -A_{21}, -A_{31}) + \\ &m_2 \cdot (-A_{12}, -A_{22}, -A_{32}) + \\ &m_3 \cdot (-A_{13}, -A_{23}, -A_{33}) = \\ &\quad (c_1, c_2, c_3) \end{aligned}$$

this is already a lattice, but let's put more columns to get m_1 :

$$\begin{aligned} &1 \cdot (c_1, c_2, c_3, 0) + \\ &m_1 \cdot (-A_{11}, -A_{21}, -A_{31}, 1) + \\ &m_2 \cdot (-A_{12}, -A_{22}, -A_{32}, 0) + \\ &m_3 \cdot (-A_{13}, -A_{23}, -A_{33}, 0) = \\ &\quad (c_1, c_2, c_3, m_1) \end{aligned}$$

this didn't work but it could've been a good idea. maybe i could make it work by using various equations so that the lattice is bigger, or some rescaling issue.

actual sol

turns out $\vec{r}$ is fixed, so for 2 ciphertexts:

$$\begin{aligned} \vec{c}_1 &= A \cdot m_1 + \vec{r} \\ \vec{c}_2 &= A \cdot m_2 + \vec{r} \end{aligned}$$

so their difference is $\vec{c}_1 - \vec{c}_2 = A \cdot (\vec{m}_1 - \vec{m}_2)$. this means $\vec{m}_1 - \vec{m}_2 = A^{-1} \cdot (\vec{c}_1 - \vec{c}_2)$

so we can gather the difference between one character and the next, for all characters. so we can bruteforce the first character (or not, we know it's H due to the flag format), and increment or decrement according to the first value of the difference. this yields

HTB{r3duc1nG_tH3_l4tTicE_l1kE_n0b0dY's_pr0bl3M}

ok my bad bro...